

HW 05

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Exercise 1

Collaborators: Sam

Tools and resources used: I used a lot of stack exchange and a little Gemini.

Exercise 2

- Points attempted: 3
- Self-assessed score: 3.
- Time spent: 3 minutes.
- Difficulty: 1/10.
- Questions/Feedback: None.

Done: For the blog post, I am looking to do something related to the topology prompt. I am currently only in a group of 2, and that is with Justin.

Exercise 3

- Points attempted: 10
- Self-assessed score: 10.
- Time spent: 60 minutes.
- Difficulty: 6/10.
- Questions/Feedback: Correctness, clarity.

Problem 7. 7.1 If G is a group of infinite order, then all subgroups of G have infinite order.

False: A counterexample would be that the group of $\{1, -1\}$ under multiplication is a subgroup of the group of non-zero rationals under multiplication.

7.2 If G is an abelian group, then all subgroups of G are abelian groups.

True: Commutativity applies to all elements of G , so by definition a subgroup of the elements of G must also be commutative.

7.3 If H is a normal subgroup of $(G, *)$, then $g * h = h * g$ for all $g \in G$ and $h \in H$.

False: The normal subgroup invariant only requires that the cosets $gH = Hg$, however, it doesn't necessarily mean that $g * h = h * g$, just that $g * h$ and $h * g$ are in the same coset $gH = Hg$.

7.4 Suppose that $(X, \star)$ is a set with a binary operation that is not necessarily a group, and there exists a "left identity" $x \in X$ such that $x \star a = a$ for all $a \in X$. Then x is also a "right identity", that is $a \star x = a$ for all $a \in X$.

False: Imagine that the binary operation $\star$ just selects the element on its right, so all elements are the "left identity" and no elements are the "right identity."

7.5 If G is a group and $g \in G$, then every column of the Cayley table for G contains g exactly once.

True: we know this to be true because for every $g' \in G$, $g * g^{-1} * g' = g'$, so that must be a column of g .

7.6 The set of rational numbers $\mathbb{Q}$ forms a group under multiplication.

False: There is no inverse of 0.

7.7 If G is a finite group of order n , then for every element $g \in G$, $g^n = e$.

True: the order m of all subgroups H must be a factor of G , so considering all subgroups generated by a single element $g \in G$, the order of $\langle g \rangle$ must be a factor of the order of G , so $g^n = e^{n/m} = e$.

Problem 8 Let G be a (not necessarily abelian) group, and let H be a (not necessarily) subgroup of G . Let the normalizer of H be defined as

$$N_G(H) = \{g \in G \mid gH = Hg\}.$$

8.1 Prove that $N_G(H) \leq G$.

Proof. We begin by noticing that $N_G(H)$ is a subset of G . Thus, the group axiom of associativity, is already satisfied.

First we show that $N_G(H)$ has an identity element. Notice $e_G \in N_G(H)$ because $e_G H = H e_G$. So, e_G is the identity element of $N_G(H)$.

Next, assuming that $n \in N_G(H)$, we can show that n^{-1} must also be in $N_G(H)$. First, we see that for every n and $h_1 \in H$, there exists a unique $h_2 \in H$, such that $n * h_1 = h_2 * n$. Now, we left and right multiply by n^{-1} , to get

$$n * h_1 = h_2 * n \tag{1}$$

$$n^{-1} * n * h * n^{-1} = n^{-1} * h_2 * n * n^{-1} \tag{2}$$

$$h * n^{-1} = n^{-1} * h_2. \tag{3}$$

Therefore, $n^{-1} \in N_G(H)$, so every element has an inverse.

Lastly, we show that $N_G(H)$ is closed. That is, for all $n_1, n_2 \in N_G(H)$, $n_1 * n_2 \in N_G(H)$ and $(n_1 * n_2)H = H(n_1 * n_2)$. Let us pick some $h_1, h'_1, h_2, h'_2 \in H$, such that

$$n_1 * h_1 = h'_1 * n_1 \quad (4)$$

$$n_2 * h_2 = h'_2 * n_2. \quad (5)$$

We isolate n_1 in equation (4) as $n_1 = h'_1 * n_1 * h_1^{-1}$. We left multiply equation (5) by n_1 to get

$$n_1 * n_2 * h_2 = h'_1 * n_1 * h_1^{-1} * h'_2 * n_2. \quad (6)$$

Now, because $n_1 H = H n_1$, there must exist some $h_1'^{-1}$ where $n_1 * h_1^{-1} = h_1'^{-1} * n_1$. We substitute this into equation (6) to get

$$= h'_1 * h_1'^{-1} * n_1 * h'_2 * n_2. \quad (7)$$

Similarly for $n_1 * h'_2$, there exists some h_2'' where $n_1 * h'_2 = h_2'' * n_1$. So, we have,

$$= h'_1 * h_1'^{-1} * h_2'' * n_1 * n_2. \quad (8)$$

Since H itself is closed, $h'_1 * h_1'^{-1} * h_2'' \in H$. Therefore, $n_1 * n_2$ is also a normalizer $N_G(H)$. Thus, $N_G(H)$ is closed, so $N_G(H)$ is a subgroup of G . $\square$

8.2 Prove that for all $g \in N_G(H)$ and $h \in H$, $ghg^{-1} \in H$.

Proof. We begin by noticing that $gH = Hg$. That is, there for every $h \in H$, there exists a unique $h' \in H$ where $g * h = h' * g$. We right multiply by g^{-1} to get $g * h * g^{-1} = h'$. Thus, for all $g \in N_G(H)$ and $h \in H$, $ghg^{-1} \in H$. $\square$

Choice Exercise 4

1. We define a normal subgroup of $(G, *)$ as one where $g * n * g^{-1} \in N$ for all $n \in N$ and $g \in G$. Do any subset of the following questions.

(a) Explain why the following are true for every subgroup $H \leq G$:

- i. For all $g \in G$ and $h \in H$, $g * h * g^{-1} \in G$

Notice that because $H \leq G$, $h \in G$. Because G is a group, it must be closed under $*$, so $g * h * g^{-1} \in G$.

- ii. For all $g \in H$ and $h \in H$, $g * h * g^{-1} \in H$

H itself is a group. Therefore it is closed under $*$. So, $g * h * g^{-1} \in H$.

(b) Prove that if G is abelian then all subgroups $H \leq G$ are normal.

Proof. We begin by noticing that because G is abelian, all subgroups H must also be abelian. Now, we construct the cosets gH, Hg for all $g \in G$. Notice that because G is abelian, $g * h = h * g$ for all $h \in H$ and $g \in G$. Therefore, $gH = Hg$. $\square$

(c) Prove that both the trivial subgroup $\{e\} \leq G$ and the group G itself are normal subgroups of G

Proof. First, we shall prove that the trivial subgroup $\{e\} \triangleleft G$. By definition of the identity element e , $g * e = e * g$ for all elements $g \in G$. Therefore, the cosets $g\{e\} = \{e\}g$, so $\{e\} \triangleleft G$.

Next, to see $G \triangleleft G$, notice that for all $g \in G$, the coset gG is just the set of G . Similarly, $Gg = G$. Therefore, $gG = Gg$ and $G \triangleleft G$. $\square$

(d) Let $G = D_6$, and recall that all elements of D_n can be written of the form $r^k f^m$. Use the fact that $r^k f = f r^{-k}$ to show that $\langle r^2 \rangle$ is a normal subgroup of D_6 .

Proof. Let $H = \langle r^2 \rangle = \{e, r^2, r^4\}$.

Now, to show that $H \triangleleft G$, we must show that for all $g \in G$, $gH = Hg$. If g is of the form r^n , $gH = Hg$ by commutativity of r .

Now, we check the case where g is of the form $r^n f^m$. Now, the cosets can be described as

$$Hg = \{r^j r^n f^m\} = \{r^{n+j} f^m\} \quad (9)$$

$$gH = \{r^n f^m r^j\} = \{r^{n-j} f^m\} \quad (10)$$

where $n \in [0, 5], m \in [0, 1], j \in \{0, 2, 4\}$.

Therefore, for any $h = r^j \in H$ and $g \in G$, there exists a unique $h' = r^{-j} \in H$ where $gH = Hg$. Thus, $H \triangleleft G$. $\square$

(e) [2] Let $G = D_n$. Show that if d is a divisor of n then $\langle r^d \rangle$ is a normal subgroup of D_n . Hint: this is a generalization of the above part, so do that part first!

Proof. In the above proof, we showed that $h = r^j$ can be mapped to $h' = r^{-j}$ in order to show that $\langle r^2 \rangle$ is a normal subgroup of D_6 . The reason this works is because 6 is a multiple of 2, so $6 - 2 = 4$ is also a multiple of 2, so $(r^j)^{-1}$ must be constructable from r^2 . This idea generalizes to any n and d . If n is a multiple of d , then its corresponding element in the right coset must be constructable via r^d . $\square$

(f) Recall that $\{\pi \in S_4 \mid \pi(4) = 4\}$ is a subgroup of S_4 . Prove that it is not a normal subgroup of S_4 .

Proof. We begin by noticing that $\{\pi \in S_4 \mid \pi(4) = 4\}$ is isomorphic to S_3 . If we can prove that S_3 is not a normal subgroup of S_4 , we also prove the proposition.

We disprove $H \triangleleft G$ through a counterexample. We can pick $g = (14) \in S_4$ and $h = (12) \in S_3$. If S_3 were a normal subgroup of S_4 , then for any $g \in S_4$ and $h \in S_3$, there exists a $h' \in S_3$ where $g \circ h = h' \circ g$. One counter example where h' is impossible to find means that S_3 is not a normal subgroup of S_4 .

We pick $g = (14) \in S_4$ and $h = (12) \in S_3$. We set up the equation $(12) \circ (14) = (14) \circ h'$ to solve for h' . Here, it makes sense to switch to word notation.

$$(12) \circ (14) = (14) \circ h' \quad (11)$$

$$2431 = (14) \circ h' \quad (12)$$

$$h' = (14) \circ 2431 \quad (13)$$

$$h' = 1432. \quad (14)$$

Notice that $h' \notin S_3$, therefore S_3 is not a normal subgroup of S_4 . $\square$

- (g) [2] Prove that if $(G, *)$ is a group and H is a subgroup. Prove that if $[H : G] = 2$ then H is a normal subgroup of G .

Proof. Notice that because $[H : G] = 2$, there must be 2 distinct cosets.

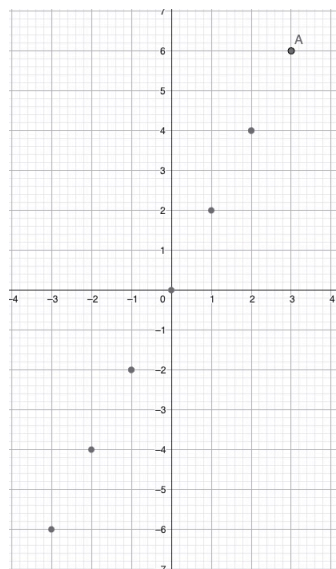
First, we look at the left cosets of H . Now, one of them must be $eH = H$ itself, and the other must be all of the other elements. In other words, gH for some $g \in G, g \notin H$. Notice that this must be the set difference $G \setminus H$.

Now, looking at the right cosets, one of them must be $He = H$. Similarly, the other coset must be the $G \setminus H$. This other coset must be of the form Hg where $g \in G, g \notin H$.

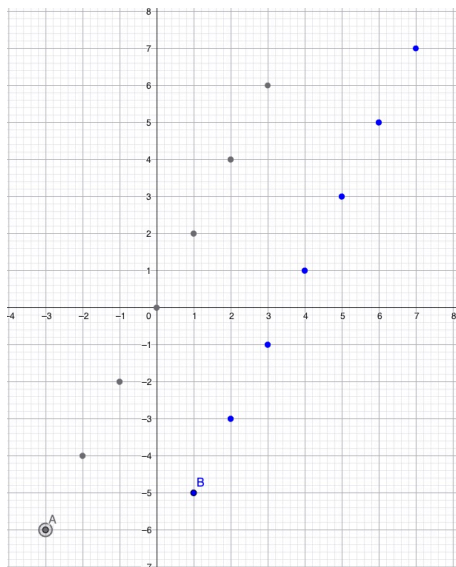
Because the left cosets partition G in the same way that the right cosets partition G , they must be equal. Therefore, $gH = Hg$ for all $g \in G$, so $H \triangleleft G$. $\square$

4. [6] Let $G = \mathbb{Z} \times \mathbb{Z}$, which we will denote $G = \mathbb{Z}^2$. Consider the subgroup $H = \langle (1, 2) \rangle$.

- (a) Plot at least four elements of H as points in the xy -plane, where $(1, 2)$.



- (b) Choose a coset $(x_1, y_1) + H$, and plot at least four elements of this coset.



- (c) Qualitatively describe the geometry of these cosets when plotted.

The two cosets are parallel.

- (d) Prove that $H \triangleleft G$.

Proof. First, notice that for $g = (a, b) \in G$, $gH = (a + h_1, b + h_2)$ for $(h_1, h_2) \in H$. By commutativity of integer addition, $Hg = (h_1 + a, h_2 + b) = (a + h_1, b + h_2) = gH$. Therefore, $H \triangleleft G$. $\square$

- (e) Prove that $G/H \cong \mathbb{Z}$ by describing an explicit isomorphism φ .

Proof. We can think of each element of G/H as the line with slope 2. Each line has a unique x-intercept, so we can define $\varphi((x, y)) = (2x - y)/2$ to be the isomorphism from G/H to $\mathbb{Z}$. $\square$